

ANTHONY BWEMBYA

Experimental Physicist | Detector Systems | Signal Processing | Scientific Computing

Netherlands | +31 6 8406 8779 | anthonybwembya@gmail.com | abwembya.com | LinkedIn | GitHub | ORCID

PROFILE

Experimental physicist with a PhD from Radboud University and more than five years of experience in detector systems, sensor calibration, scientific instrumentation, signal processing, algorithm development and experimental data analysis. My doctoral work within the Pierre Auger Observatory focused on the AugerPrime Radio Detector and on radio interferometric reconstruction methods for a distributed sensor array covering 3,000 km², leading to the first successful reconstruction of a very inclined AugerPrime air shower. I combine hands on detector work with Python and C++ software, simulation, numerical optimisation, uncertainty analysis and system validation. Additional experience covers radiation detection, LiDAR and fluorescence camera calibration, PMTs, CVD diamond sensors, VNA and DAC/AWG instrumentation, teaching, and field operation of complex experimental systems.

CORE COMPETENCIES

- **Detector and sensor systems:** characterisation, calibration, instrumentation and system validation for complex multi sensor experiments
- **Signal processing and RF sensing:** radio interferometry, sensor fusion, antenna response, waveform analysis and timing synchronisation
- **Software and data:** scientific Python and C++, automated measurement workflows, data quality diagnostics and machine learning
- **Modelling and statistics:** simulation, numerical optimisation, uncertainty propagation and model to data comparison
- **Collaboration and communication:** international teams of physicists, engineers and technicians; teaching, supervision, and writing for specialist and public audiences

PROFESSIONAL AND RESEARCH EXPERIENCE

PhD Researcher, Experimental Detector Physics and Radio Interferometry

2021 to 2026

Radboud University / Pierre Auger Observatory, Netherlands and Argentina

- Developed precision timing and calibration methods for the AugerPrime Radio Detector, a distributed antenna array covering 3,000 km² and operating in demanding environmental conditions.
- Developed and validated an automated timing calibration workflow using stable radio transmitters and independent validation signals. In a representative event, reduced failing triangle closure tests from 334 to zero and improved the average closure residual from approximately 25.4 ns to 0.18 ns.
- Developed radio interferometric algorithms for the first successful reconstruction of a very inclined AugerPrime air shower, applying sensor fusion to antenna signals under timing, geometry, noise and atmospheric constraints.
- Built reproducible Python and C++ pipelines for signal extraction, phase analysis, detector quality diagnostics, noise characterisation, simulation and reconstruction, run on HPC clusters with SLURM.
- Applied numerical optimisation to improve the accuracy of the reconstruction and to resolve ambiguities in the calibration.
- Investigated antenna response, detector geometry, atmospheric refractivity, timing uncertainty and amplitude systematics through controlled simulation and sensitivity studies.
- Supported LiDAR and fluorescence camera calibration and operations for atmospheric monitoring and detector performance assessment; used differential GPS measurements for detector position and orientation surveying.
- Worked with scientists, engineers, software developers and field technicians across an international collaboration; documented algorithms and validation procedures and presented results at collaboration meetings and workshops.

Teaching Assistant

2021 to 2025

Radboud University, Netherlands

- Supported courses in machine learning in particle physics, experimental particle physics, subatomic physics and academic writing.
- Supervised a student visit to the Boulby Underground Laboratory (2023 to 2024).

Radiation Protection Officer

2021 to 2022

Radiation Protection Authority of Zambia

- Performed radiation protection assessments and technical investigations using radiation measurement data, calibration records and uncertainty aware interpretation.

- Reviewed detector calibration and measurement traceability for laboratories and regulated organisations and prepared technical reports and recommendations.

Research and Laboratory Intern

National Institute for Scientific and Industrial Research, Zambia

- Worked with alpha and gamma spectroscopy, X ray fluorescence, detector based measurements, calibration, sample preparation and non destructive radiation analysis.

Physics Tutor

2016 to 2017

University of Zambia

- Tutored undergraduate physics students alongside completing my BSc.

SELECTED INSTRUMENTATION AND DETECTOR EXPERIENCE

- **AugerPrime Radio Detector:** distributed radio sensor array, timing calibration, antenna response, signal processing and interferometric reconstruction
- **Belle II / JENNIFER2:** CVD diamond sensor work connected with LUMIBELLE2 detector instrumentation at KEK
- **Optical and atmospheric instrumentation:** LiDAR operation, fluorescence camera calibration and PMT calibration; ICTP course in LiDAR and optics
- **Laboratory tools:** VNA, DAC/AWG, radiation detectors, spectroscopy systems and IV/CV related measurement familiarity
- **Field systems:** detector commissioning, remote monitoring, sensor diagnostics and measurements in international experimental environments

OUTREACH, PROGRAMMES AND PROJECTS

- **Nuclear Education Ambassador programme (first cohort, nearly three years):** media, journalism and non specialist writing training; produced write ups for international and government conferences on nuclear science and technology. Nuclear Ambassador for Rosatom, Moscow (2019 to 2021).
- **SKAO Development in Africa with Radio Astronomy, DARA (2017 to 2018):** astronomy training alongside school visits, public science activities and media appearances.
- **FIAS Scientific Outreach Summer School, Germany:** training in communicating science to the public.
- **Funded project:** satellite, GIS and IoT water level monitoring for Zambian energy security (about GBP 5,000).

EDUCATION

PhD in Experimental Physics

2021 to 2026

Radboud University, Netherlands

- Thesis: *In Phase With the Cosmos: Mass Composition of Cosmic Rays via Radio Interferometry*. PhD awarded 6 October 2026. [Radboud Repository](#) | Defence records: [Radboud University](#) | [Nikhef](#)

MSc in Nuclear Physics and Technology

2018 to 2021

National Research Nuclear University MEPhI, Russia

- Department of Experimental Nuclear Physics and Cosmophysics; GALPROP modelling and PAMELA / AMS 02 data analysis of the cosmic ray positron excess, supervised by Prof. Vladimir Mikhailov. Funded under the Zamatom initiative.

BSc in Physics

2017

University of Zambia

TECHNICAL TOOLKIT

- **Programming and computing:** Python, C++, Bash, Linux, Git, HPC and SLURM
- **Scientific libraries:** NumPy, SciPy, pandas, ROOT, Jupyter, scikit learn, PyTorch
- **Simulation:** CORSIKA, CoREAS, ZHAireS, GALPROP; familiarity with AIRES
- **Methods:** signal processing, Fourier and spectral analysis, optimisation, uncertainty propagation, machine learning (classification, regression, neural networks, anomaly detection, model evaluation)

PUBLICATIONS AND PRESENTATIONS

Publications: full list on ORCID, [0000-0002-6995-5978](#).

Presentations: results presented at Pierre Auger Collaboration meetings and workshops; poster presentation at a Netherlands physics conference; public engagement through Nikhef Open Days.

LANGUAGES AND AVAILABILITY

Languages: English fluent and preferred | Lozi, Bemba, Nyanja native | Russian working proficiency | Dutch developing.

Availability: holds a Netherlands research visa, no visa sponsorship required; open to relocation and international travel.